

Why MEM Works: Answer-Space Bias in Wordle

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Abstract

Modified Entropy Maximization (MEM) separates the entropy-based Wordle strategy into two components: diversity, which measures how many feedback patterns are generated, and uniformity, which measures how evenly the remaining candidates are divided. Empirical results show that Wordle is solved more efficiently when the diversity term is emphasized, for example around $\lambda = 0.7$, rather than at the entropy-equivalent value $\lambda = 0.5$. This short essay argues that this phenomenon is caused by the structural bias of the Wordle answer space. To support this interpretation, MEM is compared with Bulls and Cows, a similar guessing game whose answer space is much more symmetric. In Bulls and Cows, $\lambda = 0.5$ gives the best performance, suggesting that the advantage of diversity-heavy MEM in Wordle is not universal, but arises from the linguistic bias of English word candidates.

1 Introduction

Wordle and Bulls and Cows are both deduction games. In each game, the player repeatedly makes a guess and receives feedback about the hidden answer. The feedback partitions the current candidate set into smaller subsets, and a good strategy should choose guesses that reduce uncertainty as efficiently as possible.

A common approach to Wordle is entropy maximization. For a guess g , the feedback patterns divide the candidate answer set S into blocks. If $S_p(g)$ denotes the set of candidates that would produce feedback pattern p against guess g , then the entropy of g is

$$H(g) = - \sum_p \frac{|S_p(g)|}{|S|} \log_2 \frac{|S_p(g)|}{|S|}.$$

A guess with high entropy is expected to reduce uncertainty by a large amount. However, entropy alone does not explain *why* a guess is good. A high entropy value can come from two different sources. First, the guess may generate many different feedback patterns. Second, the resulting blocks may be close to uniform in size. Modified Entropy Maximization (MEM) separates these two effects.

2 Modified Entropy Maximization

MEM decomposes the quality of a guess into two scores: a diversity score and a uniformity score. Let $k(g)$ be the number of nonempty feedback blocks generated by guess g . The diversity score measures how many distinct feedback outcomes are actually used:

$$B(g) = \frac{\log_2 k(g)}{\log_2 k_{\max}},$$

where $k_{\max}$ is the maximum possible number of nonempty blocks under the current candidate set and the feedback rule.

The uniformity score measures how evenly the candidate set is divided among the nonempty blocks:

$$U(g) = \frac{H(g)}{\log_2 k(g)}.$$

Here, $U(g)$ is close to 1 when the nonempty blocks are nearly equal in size. Using these two components, MEM defines the score

$$M_\lambda(g) = B(g)^{2\lambda} U(g)^{2(1-\lambda)}, \quad 0 \leq \lambda \leq 1.$$

The parameter λ controls the relative importance of diversity and uniformity. When $\lambda = 0.5$, we get

$$M_{0.5}(g) = B(g)U(g) = \frac{H(g)}{\log_2 k_{\max}},$$

which is the normalized entropy. Thus $\lambda = 0.5$ corresponds to the usual entropy-based strategy. If $\lambda > 0.5$, MEM emphasizes diversity more strongly. If $\lambda < 0.5$, MEM emphasizes uniformity more strongly.

3 Wordle and Linguistic Bias

In Wordle, empirical results show that the entropy-equivalent value $\lambda = 0.5$ is not the best choice. Instead, a diversity-heavy value such as $\lambda = 0.7$ gives better practical performance. This suggests that, in Wordle, generating many different feedback patterns is more valuable than making the partition perfectly uniform.

The reason is that the Wordle answer space is not a uniform collection of arbitrary five-letter strings. It is a small set of English words. Therefore, the answer candidates have strong linguistic structure. Some letters appear more often than others; vowels and consonants follow nonuniform positional patterns; certain letter combinations are common, while others almost never occur. For example, English words contain patterns such as common vowel placements, common consonant clusters, and frequent endings. These structures create a biased candidate space.

Because of this bias, a guess that creates many distinct feedback patterns may be especially useful. Such a guess can detect several different structural features of the hidden word at once. Even if the resulting partition is not perfectly uniform, the guess may expose important distinctions among candidates. Thus, in Wordle, diversity is not merely a combinatorial property of the feedback rule. It is a way of interacting with the biased geometry of English words.

This explains why $\lambda = 0.7$ can outperform $\lambda = 0.5$ in Wordle. The value $\lambda = 0.5$ treats diversity and uniformity in the entropy-balanced way. However, Wordle rewards a strategy that slightly favors diversity, because the answer space itself is already highly structured.

4 Comparison with Bulls and Cows

To test whether the diversity-heavy behavior is specific to Wordle, we compare it with Bulls and Cows. Bulls and Cows is a similar guessing game, but its answer space is much more symmetric. In the classical version, each answer is an n -digit code with no repeated digits. In the experiment below, leading zero is allowed, so the $n = 3$ answer space has

$$10P3 = 10 \times 9 \times 8 = 720$$

possible answers. This makes the answer space close to a uniform combinatorial object rather than a linguistically biased set.

The same MEM strategy was tested for $\lambda = 0.0, 0.1, \dots, 1.0$. The results for $n = 3$ are shown in Table 1.

λ	First guess	Mean attempts	Max attempts
0.0	012	8.2972	11
0.1	012	5.9222	8
0.2	012	5.2306	7
0.3	012	5.1597	7
0.4	012	5.1236	7
0.5	012	5.0250	7
0.6	012	5.0333	7
0.7	012	5.0333	7
0.8	012	5.0319	7
0.9	012	5.0333	7
1.0	012	5.0319	7

Table 1: MEM performance on Bulls and Cows with $n = 3$.

The best mean performance occurs at $\lambda = 0.5$. This is exactly the entropy-equivalent setting. Increasing λ beyond 0.5 does not improve the result, unlike in Wordle.

This contrast is important. If diversity-heavy MEM were always superior, then we would expect $\lambda > 0.5$ to also perform better in Bulls and Cows. However, the experiment shows that this is not the case. In a more symmetric answer space, the entropy-balanced strategy remains optimal. Therefore, the advantage of $\lambda = 0.7$ in Wordle should not be interpreted as a universal property of the MEM formula. Rather, it reflects a property of the Wordle answer space.

5 Interpretation

The comparison suggests that the optimal value of λ reflects the structure of the answer space. In a symmetric game such as Bulls and Cows, each candidate is generated by a simple combinatorial rule. There is no natural-language frequency, no preferred digit pattern, and no hidden linguistic constraint. Thus, the best strategy is to divide the candidate space in an entropy-balanced way. This corresponds to $\lambda = 0.5$.

In Wordle, the situation is different. The candidates are English words, and English words are far from uniformly distributed among all possible five-letter strings. The candidate space contains linguistic bias. Therefore, a good guess should not only split the remaining candidates evenly, but also reveal as many structural distinctions as possible. This makes diversity more valuable, shifting the best value of λ above 0.5.

In this sense, λ can be interpreted as more than a hyperparameter. It can be viewed as an indicator of answer-space bias. When the answer space is symmetric and uniform, the optimal value should be close to 0.5. When the answer space is biased or structured, the optimal value may move away from 0.5. For Wordle, the empirical success of $\lambda = 0.7$ suggests that its linguistic structure favors diversity-dominant partitioning.

6 Conclusion

MEM works because it separates two components that are mixed together in entropy: diversity and uniformity. The comparison between Wordle and Bulls and Cows shows that the balance between these two components depends on the structure of the answer space. In Bulls and Cows, where the answer space is symmetric, $\lambda = 0.5$ gives the best result. In Wordle, where the answer space is biased by English word structure, a diversity-heavy value such as $\lambda = 0.7$ performs better.

Therefore, the success of MEM in Wordle is not simply due to replacing entropy with another scoring formula. Its real value is that it reveals how the geometry of the answer space affects the optimal guessing strategy. Wordle is not solved best by perfectly balanced partitions alone. It is solved better by partitions that expose the biased structure of English words.